

OPTRONIC SENSORS

For Imaging



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Two most common types of imaging sensors with potential military applications include charge-coupled device (CCD) sensors and complementary metal-oxide semiconductor (CMOS) sensors. Ladar sensor, which employs a two-dimensional array of avalanche photodiodes, is another important imaging sensor that finds application in precision-guided weapon seeker systems.

Both CCD and CMOS types of sensors use a two-dimensional array of thousands to millions of discrete pixels. The amount of light falling on each of the pixels generates free electrons with the number of electrons and hence the quantum of charge depending upon the intensity of impinging photons. These sensors differ in the mode in which this charge is converted into voltage and subsequently read out of the chip for further processing.

Charge-coupled device

Charge-coupled devices (CCDs) are basically an array of thousands to millions of light-sensitive elements called pixels etched onto a silicon surface. Each of the pixels is a buried-channel MOS capacitor.

CCDs are typically fabricated on a p-type substrate and a buried channel is implemented by forming a thin n-type region on its surface. A silicon dioxide layer is grown on top of the n-type region and an electrode, also called gate, on top of the insulating silicon dioxide completes the MOS capacitor. The electrode could be metal, but is more likely to be a heavily doped polycrystalline silicon conducting layer (Fig. 1). The sensor is not actually flat, but has

tiny cavities (like wells) that trap the incoming light and allows it to be measured. Each of these wells or cavities is a pixel.

The size of CCDs is specified in megapixels. Megapixel value can be computed by multiplying number of pixels in a row by number of pixels in a column. For example, 1000 pixels in a row and 1000 pixels in a column make 1.0-megapixel CCD chip.

When the light reflected off the target to be imaged is incident on the array of pixels, the impinging photons generate free electrons in the region underneath the pixels. In order to make sure that these free electrons don't combine with holes and disappear as heat energy, the electrons underneath each pixel are held there by applying a positive bias voltage to the pixels.

If the sensor array were exposed to light for the same time, the number of electrons and hence the quantum of charge held under a certain pixel would vary directly as per the luminous intensity that particular pixel is exposed to. This charge pattern represents the light pattern falling on the device. The charge is read out by suitable electronics and then converted into a digital bit pattern that can be understood and stored in a

SOME MILITARY SYSTEMS USING IMAGING SENSORS AND THEIR APPLICATIONS

Military systems using imaging sensors	Applications
Thermal imaging cameras	Law enforcement, marine safety and other surveillance related applications
Military satellites, space vehicles and missile seekers	Earth observation imaging and military video rate applications
Directed energy weapon systems	Target detection and tracking
Electronic and electro-optic counter-measure systems	Target detection and imaging
Night vision devices	Low-light-level imaging
Sensor systems	Space science imaging to capture detailed images across ultraviolet, visible and near-infrared spectrum
Ladar sensors	3D imaging of targets
Hyperspectral imaging sensor systems	To look at many spectra of light in closely spaced bandwidths to allow not only the presence of a target but also its features such as the material it is made of and the kind of paint it has

Imaging sensors are an important component of a wide range of defence systems and are projected to play a growing role in the coming years. This article discusses CCD, CMOS and Ladar sensors in detail